Evaluation runs: 2pc-lb (two-pc-lbinria)

generated 2026-11-08 from brain-data 522afaee3 2026-11-08 12:20:57 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	15	76	conforms	0.0	2	1 h, 250 GB	7	36	brain03, 2026-11-02
3	54	1690	conforms	0.1	2	1 h, 250 GB	15	284	brain03, 2026-11-02
4	233	72346	conforms	1.6	2	1 h, 250 GB	62	8003	brain03, 2026-11-02
5	1198	5234244	conforms	43.7	28	48 h, 250 GB	65	24615	brain03, 2026-11-07
6	7253	576552418	conforms	6843.0	614	48 h, 250 GB	386	1611773	brain05, 2026-11-07

stateless: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	15	60	12	violation	0.0	2	1 h, 250 GB	19	216	brain03, 2026-11-02
3	54	826	324	violation	0.7	4	1 h, 250 GB	339	17694	brain03, 2026-11-02
4	233	16618	9072	violation	44.4	6	1 h, 250 GB	9136	2121572	brain03, 2026-11-02
5	1198	447444	291600	violation	3150.1	12	1 h, 250 GB	291763	349481933	brain03, 2026-11-02
6	7253			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 6 bag counting-bug, 6 bag correct, 6 correct bag, 6 bag counting-bug.t16.p.r3, 7 bag correct, 7 bag counting-bug, 7 bag counting-bug.t16.p.r3, 7 bag correct.t16.p.r3, 8 bag counting-bug.

left out by the table rules (ok): 5 correct bag.

left out by the table rules (timeout): 6 correct bag.

skipped: a smaller size had reached its limit: 7 counting-bug bag, 7 correct bag, 8 correct bag, 8 counting-bug bag, 9 counting-bug bag, 9 correct bag, 10 correct bag, 10 counting-bug bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	15	76	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3	54	1690	conforms	0.2	2	1 h, 250 GB	brain02, 2026-11-02
4	233	72346	conforms	1.8	2	1 h, 250 GB	brain02, 2026-11-02
5	1198	5234244	conforms	46.7	29	48 h, 250 GB	brain03, 2026-11-07
6	7253	576552418	conforms	5140.0	652	48 h, 250 GB	brain04, 2026-11-05

stateful: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	15	60	12	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
3	54	826	324	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
4	233	16618	9072	violation	0.8	2	1 h, 250 GB	brain02, 2026-11-02
5	1198	447444	291600	violation	6.7	2	1 h, 250 GB	brain02, 2026-11-02
6	7253	15089218	10886400	violation	142.7	40	48 h, 250 GB	brain04, 2026-11-07
7	50900	610272436	468659520	violation	7021.0	674	48 h, 250 GB	brain05, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 6 correct bag, 7 bag counting-bug.t16.p.r3, 7 bag correct.t16.p.r3, 8 correct bag, 8 bag correct, 9 counting-bug bag, 9 correct bag, 10 correct bag, 10 counting-bug bag, 10 bag counting-bug, 11 correct bag, 11 counting-bug bag, 12 correct bag, 12 counting-bug bag, 13 correct bag, 13 counting-bug bag, 14 counting-bug bag.

left out by the table rules (ok): 5 bag correct, 5 correct bag, 6 counting-bug bag.

left out by the table rules (timeout): 6 correct bag, 7 counting-bug bag, 8 bag counting-bug.

skipped: a smaller size had reached its limit: 7 correct bag, 8 correct bag, 8 counting-bug bag, 9 counting-bug bag, 9 correct bag, 10 correct bag, 10 counting-bug bag.

the tool ended with an error (exit-101): 7 bag correct.

left out by the table rules (invalid: foreign load): 9 bag counting-bug, 9 bag correct.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	15	76	conforms	0.2	8	10 min, 16 GB	0.0	0.2	0.0	in-memory	brain05, 2026-11-02
3	54	1690	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
4	233	72346	conforms	1.7	8	1 h, 250 GB	0.1	1.7	0.0	merge	brain05, 2026-11-03
5	1198	5234244	conforms	52.0	28	48 h, 250 GB	0.1	52.0	0.0	merge	brain03, 2026-11-07
5	1198	5234244	conforms	53.5	30	1 h, 250 GB	0.1	53.5	0.0	merge	brain05, 2026-11-03
6	7253	576552418	conforms	5706.0	632	48 h, 250 GB	0.3	5706.0	0.0	merge	brain04, 2026-11-06

offline: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	15	60	4	violation	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	54	826	6	violation	0.3	8	10 min, 16 GB	0.0	0.3	0.0	in-memory	brain05, 2026-11-02
4	233	16618	32	violation	0.9	8	10 min, 16 GB	0.1	0.9	0.0	in-memory	brain05, 2026-11-02
5	1198	447444	50	violation	8.9	8	1 h, 250 GB	0.1	8.8	0.0	merge	brain05, 2026-11-03
6	7253	15089218	232	violation	142.1	38	48 h, 250 GB	0.3	142.0	0.0	merge	brain04, 2026-11-07
6	7253	15089218	232	violation	162.4	39	1 h, 250 GB	0.5	162.4	0.0	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 7 bag correct.p.r3, 7 bag counting-bug.p.r3, 8 bag counting-bug.

skipped: a smaller size had reached its limit: 6 correct bag, 7 counting-bug bag, 7 correct bag, 7 bag counting-bug, 7 bag correct, 8 correct bag, 8 counting-bug bag, 9 counting-bug bag, 9 correct bag, 10 correct bag, 10 counting-bug bag.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	15	76	conforms	3.1	162	10 min, 16 GB	3.1	0.2	0.0	in-memory	brain01, 2026-11-02
3	54	1690	conforms	4.1	172	10 min, 250 GB	4.0	0.1	0.1	merge	brain01, 2026-11-03
4	233	72346	conforms	4.4	185	10 min, 250 GB	4.4	1.7	0.0	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	15	60	4	violation	3.1	162	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3	54	826	6	violation	4.2	172	10 min, 250 GB	4.0	0.3	0.2	merge	brain01, 2026-11-03
4	233	16618	32	violation	4.5	185	10 min, 250 GB	4.4	0.9	0.1	merge	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag counting-bug, 6 bag correct, 6 bag counting-bug, 7 bag counting-bug, 7 bag correct, 8 bag counting-bug.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	15	76	conforms	3.1	158	1 h, 250 GB		brain01, 2026-11-02
3	54	1690	conforms	3.2	161	1 h, 250 GB		brain01, 2026-11-02
4	233	72346	conforms	6.0	170	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	15	60	12	violation	3.1	158	1 h, 250 GB	2	brain01, 2026-11-02
3	54	826	324	violation	3.2	161	1 h, 250 GB	2	brain01, 2026-11-02
4	233	16618	9072	violation	4.6	170	1 h, 250 GB	14	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 6 bag correct, 6 bag counting-bug, 7 bag counting-bug, 7 bag correct, 8 bag counting-bug.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	76	76	conforms	30.3	1892	4 h, 250 GB	30.3	0.2	0.0	merge	brain01, 2026-11-03
3	1690	1690	conforms	9.1	541	10 min, 250 GB	9.1	0.1	0.1	merge	brain01, 2026-11-03
4	72346	72346	conforms	225.9	3049	10 min, 250 GB	225.8	1.7	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **counting-bug**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	60	60	conforms	12.1	1684	4 h, 250 GB	12.0	0.0	0.0	merge	brain01, 2026-11-03
3	826	826	conforms	6.8	345	10 min, 250 GB	6.7	0.3	0.1	merge	brain01, 2026-11-03
4	16618	16618	conforms	92.2	2420	10 min, 250 GB	92.2	0.9	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	76	60	4	violation	30.3	1892	4 h, 250 GB	30.3	0.0	0.0	merge	brain01, 2026-11-03
3	1690	826	6	violation	9.1	541	10 min, 250 GB	9.1	0.3	0.0	merge	brain01, 2026-11-03
4	72346	16618	32	violation	225.9	3049	10 min, 250 GB	225.8	0.9	0.1	merge	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag counting-bug, 6 bag counting-bug, 6 bag correct, 7 bag counting-bug.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	76	76	conforms	30.3	1892	4 h, 250 GB	0.2	30.3	0.0	merge	brain01, 2026-11-03
3	1690	1690	conforms	9.4	541	10 min, 250 GB	0.1	9.1	0.3	merge	brain01, 2026-11-03
4	72346	72346	conforms	225.8	3049	10 min, 250 GB	1.7	225.8	0.1	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **counting-bug**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	60	60	conforms	12.0	1684	4 h, 250 GB	0.0	12.0	0.0	merge	brain01, 2026-11-03
3	826	826	conforms	6.7	345	10 min, 250 GB	0.3	6.7	0.1	merge	brain01, 2026-11-03
4	16618	16618	conforms	92.3	2420	10 min, 250 GB	0.9	92.2	0.1	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	76	60	4	violation	12.0	1684	4 h, 250 GB	0.2	12.0	0.0	merge	brain01, 2026-11-03
3	1690	826	6	violation	6.8	345	10 min, 250 GB	0.1	6.7	0.1	merge	brain01, 2026-11-03
4	72346	16618	32	violation	92.4	2420	10 min, 250 GB	1.7	92.2	0.2	merge	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag counting-bug, 6 bag counting-bug, 6 bag correct, 7 bag counting-bug.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	76	76	conforms	4.8	164	4 h, 250 GB	4.8	0.2	0.0	merge	brain01, 2026-11-03
3	1690	1690	conforms	11.6	2189	10 min, 250 GB	11.5	0.1	0.1	merge	brain01, 2026-11-03
4	72346	72346	conforms	276.6	3147	10 min, 250 GB	276.5	1.7	0.0	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **counting-bug**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	60	60	conforms	5.1	168	4 h, 250 GB	5.1	0.0	0.0	merge	brain01, 2026-11-03
3	826	826	conforms	8.1	325	10 min, 250 GB	7.9	0.3	0.2	merge	brain01, 2026-11-03
4	16618	16618	conforms	91.7	2560	10 min, 250 GB	91.7	0.9	0.0	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	76	60	4	violation	4.8	164	4 h, 250 GB	4.8	0.0	0.0	merge	brain01, 2026-11-03
3	1690	826	6	violation	12.1	2190	10 min, 250 GB	11.5	0.3	0.6	merge	brain01, 2026-11-03
4	72346	16618	32	violation	276.6	3147	10 min, 250 GB	276.5	0.9	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag counting-bug, 6 bag counting-bug, 6 bag correct, 7 bag counting-bug.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	76	76	conforms	4.8	164	4 h, 250 GB	0.2	4.8	0.0	merge	brain01, 2026-11-05
3	1690	1690	conforms	11.8	2189	10 min, 250 GB	0.1	11.5	0.3	merge	brain01, 2026-11-03
4	72346	72346	conforms	276.9	3147	10 min, 250 GB	1.7	276.5	0.3	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **counting-bug**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	60	60	conforms	5.1	168	4 h, 250 GB	0.0	5.1	0.0	merge	brain01, 2026-11-05
3	826	826	conforms	7.9	325	10 min, 250 GB	0.3	7.9	0.0	merge	brain01, 2026-11-03
4	16618	16618	conforms	92.1	2560	10 min, 250 GB	0.9	91.7	0.4	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	76	60	4	violation	5.1	168	4 h, 250 GB	0.2	5.1	0.0	merge	brain01, 2026-11-05
3	1690	826	6	violation	8.3	325	10 min, 250 GB	0.1	7.9	0.4	merge	brain01, 2026-11-03
4	72346	16618	32	violation	91.7	2560	10 min, 250 GB	1.7	91.7	0.0	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 5 bag counting-bug, 6 bag correct, 6 bag counting-bug, 7 bag counting-bug.
refused (an input missing at run time): 5 bag correct.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	76	76	conforms	3.5	169	1 h, 250 GB		brain01, 2026-11-02
3	1690	1690	conforms	15.3	1861	1 h, 250 GB		brain01, 2026-11-03
4	72346	72346	conforms	324.2	3141	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **counting-bug**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	60	60	conforms	3.4	167	1 h, 250 GB		brain01, 2026-11-02
3	826	826	conforms	12.0	1890	1 h, 250 GB		brain01, 2026-11-03
4	16618	16618	conforms	81.7	2623	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **counting-bug**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	76	60	12	violation	3.5	158	1 h, 250 GB	2069	brain01, 2026-11-02
3	1690	826	324	violation	11.7	1859	1 h, 250 GB	5063	brain01, 2026-11-03
4	72346	16618	9072	violation	83.1	2576	1 h, 250 GB	5445	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 6 bag counting-bug, 6 bag correct, 7 bag counting-bug.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	76	76	conforms	3.6	158	1 h, 250 GB		brain01, 2026-11-02
3	1690	1690	conforms	8.7	636	1 h, 250 GB		brain01, 2026-11-02
4	72346	72346	conforms	288.3	3427	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **counting-bug**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	60	60	conforms	3.1	158	1 h, 250 GB		brain01, 2026-11-02
3	826	826	conforms	5.9	392	1 h, 250 GB		brain01, 2026-11-02
4	16618	16618	conforms	64.1	2325	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair counting-bug, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	76	60	12	violation	3.4	158	1 h, 250 GB	2	brain01, 2026-11-02
3	1690	826	324	violation	8.6	636	1 h, 250 GB	3	brain01, 2026-11-02
4	72346	16618	9072	violation	287.6	3427	1 h, 250 GB	7	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 6 bag counting-bug, 6 bag correct, 7 bag counting-bug.